Thm (Euler's formula)

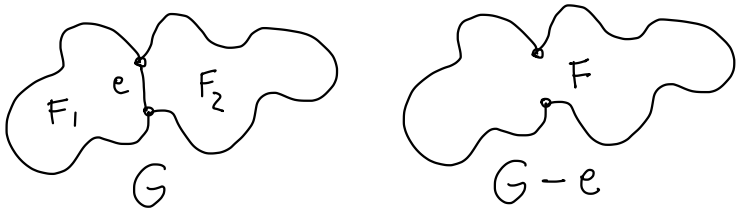
G connected plane graph $\Rightarrow$ $v(G) - e(G) + f(G) = 2$.

proof: Let $n = v(G)$. Induction on $e(G)$.

Basis of induction: G tree $\Rightarrow$ $e(G) = n-1$, $\#\text{faces} = 1$

$$n - (n-1) + 1 = 2.$$

Suppose G is not a tree. Let C be a cycle in G and $e \in E(C)$.
$\exists$ faces F_1, F_2 (distinct) such that $e \in E(G[F_1]) \cap E(G[F_2])$.



$$v(G-e) - e(G-e) + f(G-e) = 2 \Rightarrow v(G) - (e(G)-1) + f(G) - 1 = 2$$

Q/ How do maximal plane graphs look like?

<u>Defn</u>: A plane graph (G,h,p) is maximal if $\forall\, xy \in E(\bar{G})$ $\nexists$ injective continuous function $h_{xy} : [0,1] \to \mathbb{R}^2$ such that $(G+xy, h \cup h_{xy}, p)$ is a plane graph.

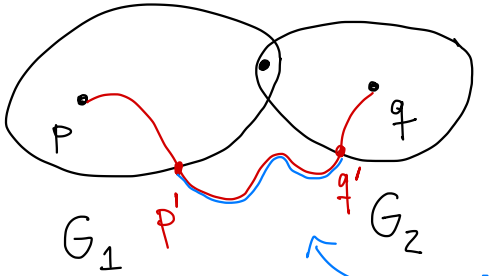
<u>Defn</u>: (G,h,p) is a plane triangulation if $G[F] \cong K_3$ $\forall$ face F.

<u>Thm</u>: G with $v(G) \geq 3$ is maximum plane $\Leftrightarrow$ G is a plane triangulation.

<u>proof</u>: $(\Leftarrow)$ Easy.

$(\Rightarrow)$ • We may assume G is connected.

Suppose for contradiction that G is not 2-connected.



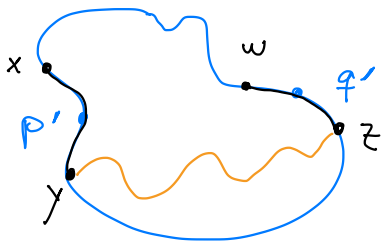
A continuous path φ between p and q.

it has to be inside a face in G.

p : point in G_1 and q : point in G_2.

$p' :=$ last point of φ in G_1, $q' :=$ first point of φ in G_2.

$xy \in E(G_1)$



$wz \in E(G_2)$.

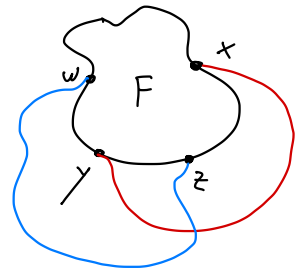
$y \neq z$.

we can link y to z! Contradiction.

G 2-connected $\Rightarrow$ $G[F]$ is a cycle $\forall$ face F.

Suppose for contradiction $\exists$ face F such that $G[F] \cong C_\ell$, $\ell \geq 4$.

Set $H = G[F]$ and take $xy \in E(\overline{H})$. Then,



contradiction!

$\blacksquare$

<u>Corollary</u>: G plane, $v(G) \geq 3 \Rightarrow e(G) \leq 3v(G) - 6$. Equality holds $\Leftrightarrow$ G is a triangulation.

<u>proof</u>:

$$2e(G) \geq \#\{(e,F) : e \in E(G[F])\} \geq 3f(G)$$

$$3(e(G) - v(G) + 2) = 3f(G) \leq 2e(G) \Rightarrow e(G) \leq 3v(G) - 6.$$

$\blacksquare$

<u>Corollary</u> : K_5 is not planar.

<u>proof</u> : $e(K_5) = 10 > 3 \cdot 5 - 6$ $\blacksquare$

Q/ What about $K_{3,3}$? $e(K_{3,3}) = 9 \geq 3 \cdot 6 - 6$.

<u>Lemma</u> : Let G be a plane graph with $e(G[F]) \geq 4$ $\forall$ face F.
Then, $e(G) \leq 2v(G) - 4$.

<u>proof</u> : $2e(G) \geq \#\{(e,F) : e \in E(G[F])\} \geq 4f(G)$

$$2(e(G) - v(G) + 2) = 2f(G) \leq e(G) \Rightarrow e(G) \leq 2v(G) - 4.$$

$\blacksquare$

<u>Corollary</u> : $K_{3,3}$ is not planar.

<u>proof</u>: Suppose for contradiction that $K_{3,3}$ is planar.

$\Delta \notin K_{3,3} \Rightarrow$ every face has ≥ 4 edges. We must have

$e(K_{3,3}) \leq 2 \cdot 6 - 4 = 8$, contradiction. $\blacksquare$

<u>Defn</u>: A polyhedron $P \subseteq \mathbb{R}^3$ is a bounded set of the form

$$P = \bigcap_{i \in [n]} \{(x,y,z) : \alpha_{i1} x + \alpha_{i2} y + \alpha_{i3} z \leq c_i\}.$$

We will always assume that this is minimal.

Let P_i be the associated plane.

An edge of P is a non-empty set of the form $P_i \cap P_j \cap P$.

A vertex is a non-empty set given by the intersection of 2 edges.

A face is the bounded open set (in 2D) in $P_j \setminus \bigcap_{i \neq j} P_i$

<u>Defn</u>: We say that $P \subseteq \mathbb{R}^3$ is a platonic solid if every vertex meets the same number of faces, and the boundary of each face is the same regular polygon.

<u>Thm</u>: There are 5 platonic solids.

<u>proof</u>: P platonic solid, G be the associated graph

<u>Notation</u>: $n = v(G)$, $m = e(G)$, r = size of cycle bounding each face

$s = d(v) \ \forall \ v \in V(G)$.

G is a planar graph! Moreover, $f \cdot r = 2m = ns$.

Euler's formula: $n - m + f = 2 \Rightarrow \frac{2m}{s} - m + \frac{2m}{r} = 2$

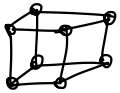
$$\Rightarrow \frac{1}{s} - \frac{1}{2} + \frac{1}{r} = \frac{1}{m} \Rightarrow \frac{1}{s} + \frac{1}{r} > \frac{1}{2}$$

- $r \geq 3 \Rightarrow \frac{1}{3} + \frac{1}{s} > \frac{1}{2} \Rightarrow s < 6$. Analogously, $r < 6$.

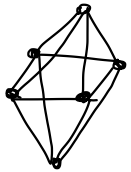
$\Rightarrow r, s \in \{3,4,5\}$.

- $r \geq 4 \Rightarrow \frac{1}{4} + \frac{1}{s} > \frac{1}{2} \Rightarrow s < 4 \Rightarrow s = 3$. Similarly, $s \geq 4 \Rightarrow r = 3$.

- $(s,r) =$ (3,3) tetrahedron

  (3,4) cube 

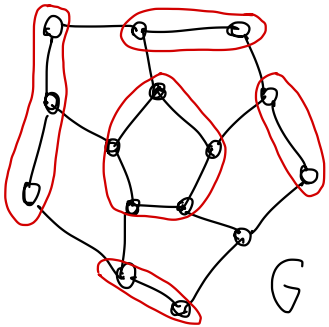
  (3,5) dodecahedron

  (4,3) Octahedron 

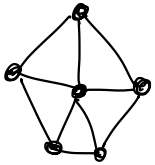
  (5,3) Icosahedron.

<u>Defn</u>: A graph X is a <u>minor</u> of a graph G if X can be obtained from G by a series of edge contractions, edge deletions and vertex deletions.

<u>Example</u>:



G



X

X is a minor of G.